

Selective expression of LDLR and VLDLR in myelinating oligodendrocytes.

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Oligodendrocytes form myelin sheaths around axons in the central nervous system. Although cholesterol is one of the major lipid components in myelin sheath, the source of cholesterol for myelination remains to be defined. In this study, we report that low-density lipoprotein receptor (LDLR) and very low-density lipoprotein receptor (VLDLR) are selectively expressed in mature myelinating oligodendrocytes in the postnatal CNS. Both receptors are specifically expressed in differentiated oligodendrocytes in P7 spinal cord, but progressively down-regulated after P15. In adult animals, only LDLR expression can be detected in a small number of oligodendrocytes throughout the entire spinal cord. In the brain region, LDLR is expressed by the white matter oligodendrocytes of both cerebellum and cerebral cortex, whereas VLDLR has a weak expression in cerebellar oligodendrocytes. Together, our expression studies suggest that cholesterol uptake by LDLR and VLDLR may play an important role in the formation of myelin sheath.

In vitro myelin formation using embryonic stem cells.

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Myelination in the central nervous system is the process by which oligodendrocytes form myelin sheaths around the axons of neurons. Myelination enables neurons to transmit information more quickly and more efficiently and allows for more complex brain functions; yet, remarkably, the underlying mechanism by which myelination occurs is still not fully understood. A reliable in vitro assay is essential to dissect oligodendrocyte and myelin biology. Hence, we developed a protocol to generate myelinating oligodendrocytes from mouse embryonic stem cells and established a myelin formation assay with embryonic stem cell-derived neurons in microfluidic devices. Myelin formation was quantified using a custom semi-automated method that is suitable for larger scale analysis. Finally, early myelination was followed in real time over several days and the results have led us to propose a new model for myelin formation.

The cell biology of CNS myelination.

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Myelination of axons in the central nervous system results from the remarkable ability of oligodendrocytes to wrap multiple axons with highly specialized membrane. Because myelin membrane grows as it ensheaths axons, cytoskeletal rearrangements that enable ensheathment must be coordinated with myelin production. Because the myelin sheaths of a single oligodendrocyte can differ in thickness and length, mechanisms that coordinate axon ensheathment with myelin growth likely operate within individual oligodendrocyte processes. Recent studies have revealed new information about how assembly and disassembly of actin filaments helps drive the leading edge of nascent myelin membrane around and along axons. Concurrently, other investigations have begun to uncover evidence of communication between axons and oligodendrocytes that can regulate myelin formation.

Expression and subcellular localization of mitochondrial docking protein, syntaphilin, in oligodendrocytes and CNS myelin sheath.

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Oligodendrocytes produce lipid-rich myelin sheaths that provide metabolic support to the underlying axon and facilitate saltatory conduction. Oligodendrocyte mitochondria supply the bulk of energy and carbon-chain backbones required for lipid synthesis. The sparsity of mitochondria in the myelin sheath suggests that tight regulation of mitochondrial trafficking is crucial for their efficient distribution in the cell. In particular, retention of mitochondria at axoglial junctions would support local lipid synthesis and membrane remodeling during myelination. How mitochondrial docking in oligodendrocytes is regulated is not known. Our findings indicate that syntaphilin (SNPH), a mitochondrial docking protein that has been characterized in neurons, is expressed by oligodendrocyte precursor cells (OPCs) and mature oligodendrocytes in vitro and present in the myelin sheath in vivo. We have previously reported that bath application of netrin-1 promotes the elaboration of myelin basic protein-positive membranes, and that localized presentation of a netrin-1 coated microbead results in rapid accumulation of mitochondria at the site of oligodendrocyte-bead adhesion. Here we show that netrin-1 increases the redistribution of SNPH to oligodendrocyte processes during the expansion of myelin basic protein-positive membranes and that SNPH clusters at the oligodendrocyte plasma membrane at sites of adhesion with netrin-1-coated beads where mitochondria are retained. These findings suggest roles for SNPH in oligodendrocytes regulating netrin-1-mediated mitochondrial docking and myelin membrane expansion.

The Rules of Attraction in Central Nervous System Myelination.

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The wrapping of myelin around axons is crucial for the development and function of the central nervous system (CNS) of vertebrates, greatly regulating the conduction of action potentials. Oligodendrocytes, the myelinating glia of the CNS, have an intrinsic tendency to wrap myelin around any permissive structure in vitro, but in vivo, myelin is targeted with remarkable specificity only to certain axons. Despite the importance of myelination, the mechanisms by which oligodendrocytes navigate a complex milieu that includes many types of cells and their cellular projections and select only certain axons for myelination remains incompletely understood. In this Mini-review, I highlight recent studies that shed light on the molecular and cellular rules governing CNS myelin targeting.

Oligodendrocyte development and myelin sheath formation are regulated by the antagonistic interaction between the Rag-Ragulator complex and TFEB.

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Myelination by oligodendrocytes is critical for fast axonal conduction and for the support and survival of neurons in the central nervous system. Recent studies have emphasized that myelination is plastic and that new myelin is formed throughout life. Nonetheless, the mechanisms that regulate the number, length, and location of myelin sheaths formed by individual oligodendrocytes are incompletely understood. Previous work showed that the lysosomal transcription factor TFEB represses myelination by oligodendrocytes and that the RagA GTPase inhibits TFEB, but the step or steps of myelination in which

TFEB plays a role have remained unclear. Here, we show that TFEB regulates oligodendrocyte differentiation and also controls the length of myelin sheaths formed by individual oligodendrocytes. In the dorsal spinal cord of *tfeb* mutants, individual oligodendrocytes produce myelin sheaths that are longer than those produced by wildtype cells. Transmission electron microscopy shows that there are more myelinated axons in the dorsal spinal cord of *tfeb* mutants than in wildtype animals, but no significant change in axon diameter. In contrast to *tfeb* mutants, oligodendrocytes in *rraga* mutants produce shorter myelin sheaths. The sheath length in *rraga; tfeb* double mutants is not significantly different from wildtype, consistent with the antagonistic interaction between RagA and TFEB. Finally, we find that the GTPase activating protein *Flcn* and the RagCa and RagCb GTPases are also necessary for myelination by oligodendrocytes. These findings demonstrate that TFEB coordinates myelin sheath length and number during myelin formation in the central nervous system.

Navigating oligodendrocyte precursor cell aging in brain health.

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Oligodendrocyte precursor cells (OPCs) comprise 5-8 % of the adult glial cell population and stand out as the most proliferative cell type in the central nervous system (CNS). OPCs are responsible for generating oligodendrocytes (OLs), the myelinating cells of the CNS. However, OPC functions decline as we age, resulting in impaired differentiation and inadequate remyelination. This review explores the cellular and molecular changes associated with OPC aging, and their impact on OPC differentiation and functionality. Furthermore, it examines the impact of OPC aging within the context of multiple sclerosis and Alzheimer's disease, both neurodegenerative conditions wherein aged OPCs exacerbate disease progression by impeding remyelination. Moreover, various pharmacological interventions targeting pathways related to senescence and differentiation are discussed as potential strategies to rejuvenate aged OPCs. Enhancing our understanding of OPC aging mechanisms holds promise for developing new therapies to improve remyelination and repair in age-related neurodegenerative disorders.

Oligodendroglial p130Cas is a target of Fyn kinase involved in process formation, cell migration and survival.

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Oligodendrocytes are the myelinating glial cells of the central nervous system. In the course of brain development, oligodendrocyte precursor cells migrate, scan the environment and differentiate into mature oligodendrocytes with multiple cellular processes which recognize and ensheath neuronal axons. During differentiation, oligodendrocytes undergo dramatic morphological changes requiring cytoskeletal rearrangements which need to be tightly regulated. The non-receptor tyrosine kinase Fyn plays a central role in oligodendrocyte differentiation and myelination. In order to improve our understanding of the role of oligodendroglial Fyn kinase, we have identified Fyn targets in these cells. Purification and mass-spectrometric analysis of tyrosine-phosphorylated proteins in response to overexpressed active Fyn in the oligodendrocyte precursor cell line Oli-neu, yielded the adaptor molecule p130Cas. We analyzed the function of this Fyn target in oligodendroglial cells and observed that reduction of p130Cas levels by siRNA affects process outgrowth, the thickness of cellular processes and migration behavior of Oli-neu cells. Furthermore, long term p130Cas reduction results in decreased cell numbers as a result of

increased apoptosis in cultured primary oligodendrocytes. Our data contribute to understanding the molecular events taking place during oligodendrocyte migration and morphological differentiation and have implications for myelin formation.

MOBP levels are regulated by Fyn kinase and affect the morphological differentiation of oligodendrocytes.

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Oligodendrocytes are the myelinating glial cells of the central nervous system (CNS). Myelin is formed by extensive wrapping of oligodendroglial processes around axonal segments, which ultimately allows a rapid saltatory conduction of action potentials within the CNS and sustains neuronal health. The non-receptor tyrosine kinase Fyn is an important signaling molecule in oligodendrocytes. It controls the morphological differentiation of oligodendrocytes and is an integrator of axon-glial signaling cascades leading to localized synthesis of myelin basic protein (MBP), which is essential for myelin formation. The abundant myelin-associated oligodendrocytic basic protein (MOBP) resembles MBP in several aspects and has also been reported to be localized as mRNA and translated in the peripheral myelin compartment. The signals initiating local MOBP synthesis are so far unknown and the cellular function of MOBP remains elusive. Here, we show, by several approaches in cultured primary oligodendrocytes, that MOBP synthesis is stimulated by Fyn activity. Moreover, we reveal a new function for MOBP in oligodendroglial morphological differentiation.

Fatty Acid Binding Protein 5 Mediates Cell Death by Psychosine Exposure through Mitochondrial Macropores Formation in Oligodendrocytes.

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Oligodendrocytes, the myelinating cells in the central nervous system (CNS), are critical for producing myelin throughout the CNS. The loss of oligodendrocytes is associated with multiple neurodegenerative disorders mediated by psychosine. However, the involvement of psychosine in the critical biochemical pathogenetic mechanism of the loss of oligodendrocytes and myelin in krabbe disease (KD) remains unclear. Here, we addressed how oligodendrocytes are induced by psychosine treatment in both KG-1C human oligodendroglial cells and mouse oligodendrocyte precursor cells. We found that fatty acid binding protein 5 (FABP5) expressed in oligodendrocytes accelerates mitochondria-induced glial death by inducing mitochondrial macropore formation through voltage-dependent anion channels (VDAC-1) and BAX. These two proteins mediate mitochondrial outer membrane permeabilization, thereby leading to the release of mitochondrial DNA and cytochrome C into the cytosol, and the activation of apoptotic caspases. Furthermore, we confirmed that the inhibition of FABP5 functions by shRNA and FABP5-specific ligands blocking mitochondrial macropore formation, thereby rescuing psychosine-induced oligodendrocyte death. Taken together, we identified FABP5 as a critical factor in mitochondrial injury associated with psychosine-induced apoptosis in oligodendrocytes.